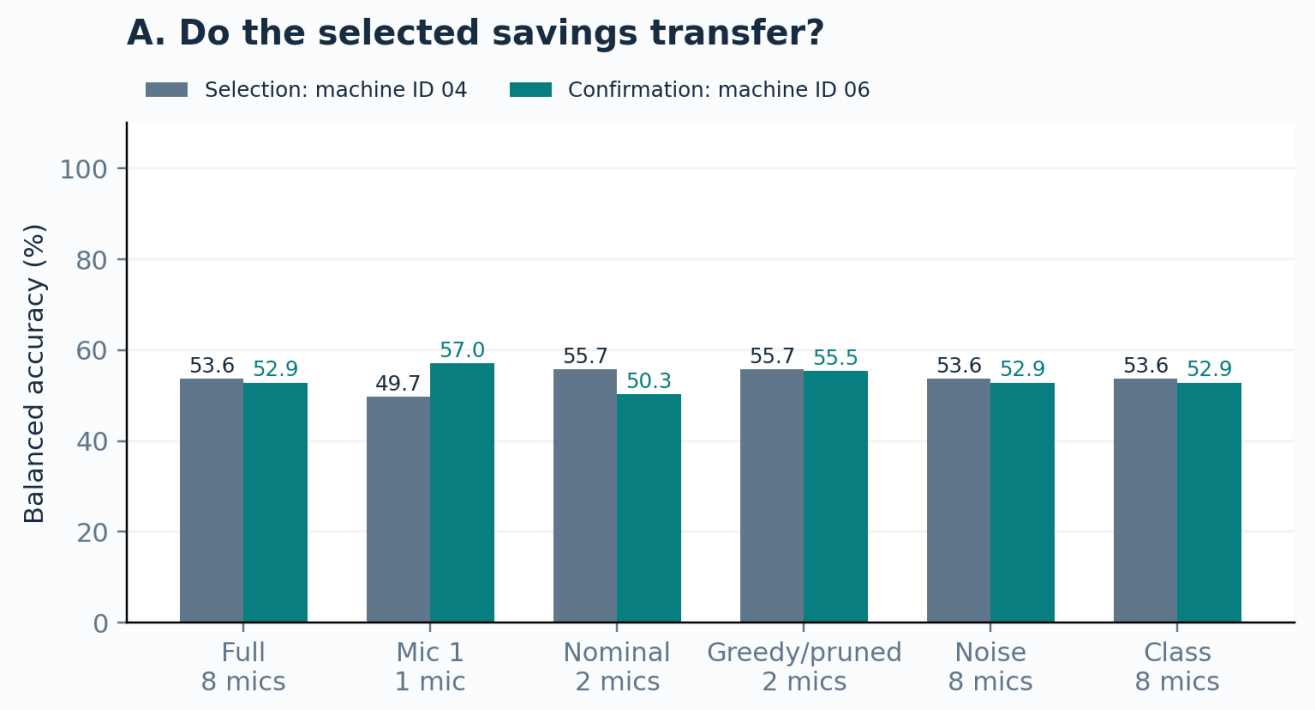
What survives when the machine changes?

Eight synchronized microphones. Every one of the 255 possible panels. Protected confirmation on different machine identities.

Demonstrated advantage. The exact finite search exposes the complete 255-panel design space, certifies every minimum channel count and identifies when requested erasure robustness is impossible. Every proposed stream reduction comes with a checkable finite optimum and its transfer limits. The predeclared greedy/pruned panel retains 2 of eight recorded streams (75% fewer) and passes the protected type-specific margin at 55.47% balanced accuracy. The exact tie choice fails. All reported panels fail the simultaneous absolute class-recall floor. This finite physical replay does not establish hardware savings or generalization to an industrial machine population.

The full array scores 52.86% balanced accuracy on the 384 protected recordings. The nominal development optimum, microphones 3, 6, scores 50.26% (-2.60 percentage points).



The primary masks were chosen on machine ID 04 and locked before prediction on ID 06. Type-specific requirements can fail even when a pooled score improves.

Exact selection and optimally pruned greedy tie on minimum channel count under every feasible objective, with matching infeasibility elsewhere. Equally cheap nominal masks differ; the complete table reports each protected outcome. Exact selection has no channel-count advantage over ordinary greedy. Both erasure objectives are infeasible.

Every comparator, under the same rules

The nominal objective protects pooled and per-type balanced accuracy. Noise-aware selection adds each type-by-noise cell. Class robustness also protects normal and abnormal recall in every such cell. Erasure objectives test every allowed surviving panel against the intact full-array thresholds.

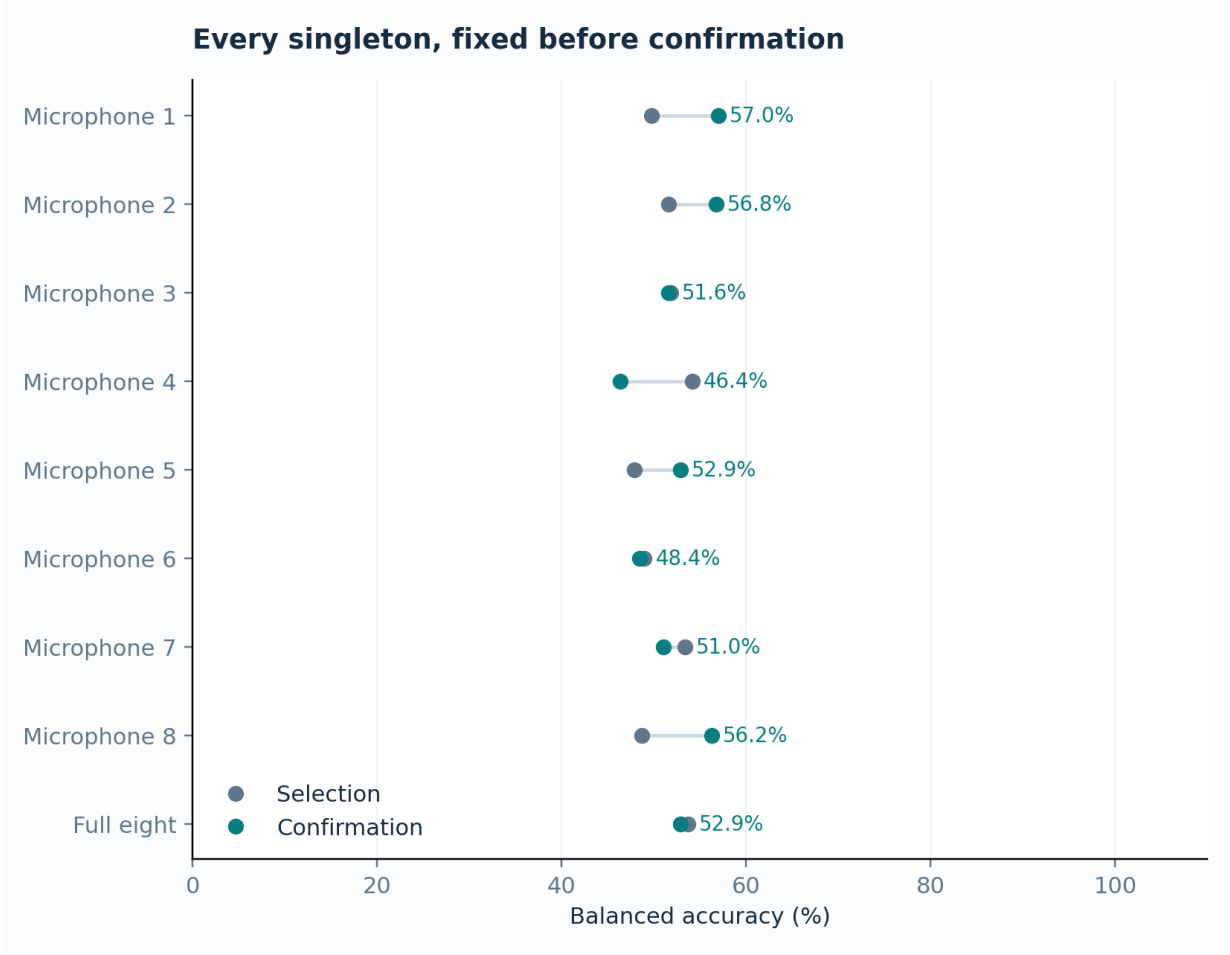
Requirement	Method	Microphones	Mics	Confirm. BA	Contract
By machine type	Exact	3, 6	2	50.26%	Fail
By machine type	Greedy	1, 7	2	55.47%	Pass
By machine type	Pruned greedy	1, 7	2	55.47%	Pass
By type + noise	Exact	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Pass
By type + noise	Greedy	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Pass
By type + noise	Pruned greedy	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Pass
By type + noise + class	Exact	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Pass
By type + noise + class	Greedy	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Pass
By type + noise + class	Pruned greedy	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Pass
Also lose any 1 mic	Exact	No panel	-	-	-
Also lose any 1 mic	Greedy*	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Fail
Also lose any 1 mic	Pruned greedy	No panel	-	-	-
Also lose any 2 mics	Exact	No panel	-	-	-
Also lose any 2 mics	Greedy*	1, 2, 3, 4, 5, 6, 7, 8	8	52.86%	Fail
Also lose any 2 mics	Pruned greedy	No panel	-	-	-

An asterisk marks a greedy terminal mask that failed selection. Pass/fail applies to the same objective on the protected confirmation table. It is a descriptive finite-panel result, not a population noninferiority claim. No channel selection is changed after confirmation. Equal-cost microphones are a declared proxy; physical price, energy and latency have not been measured.

The sample granularity matters

Each type-by-noise cell has sixteen normal and sixteen abnormal examples. One additional mistake changes balanced accuracy by 3.125 percentage points and class recall by 6.25 points. A two-point cell margin therefore permits no additional error. Relative preservation is also separate from an absolute recall floor: matching a weak full-array reference is not sufficient fault detection.

The useful channels depend on context

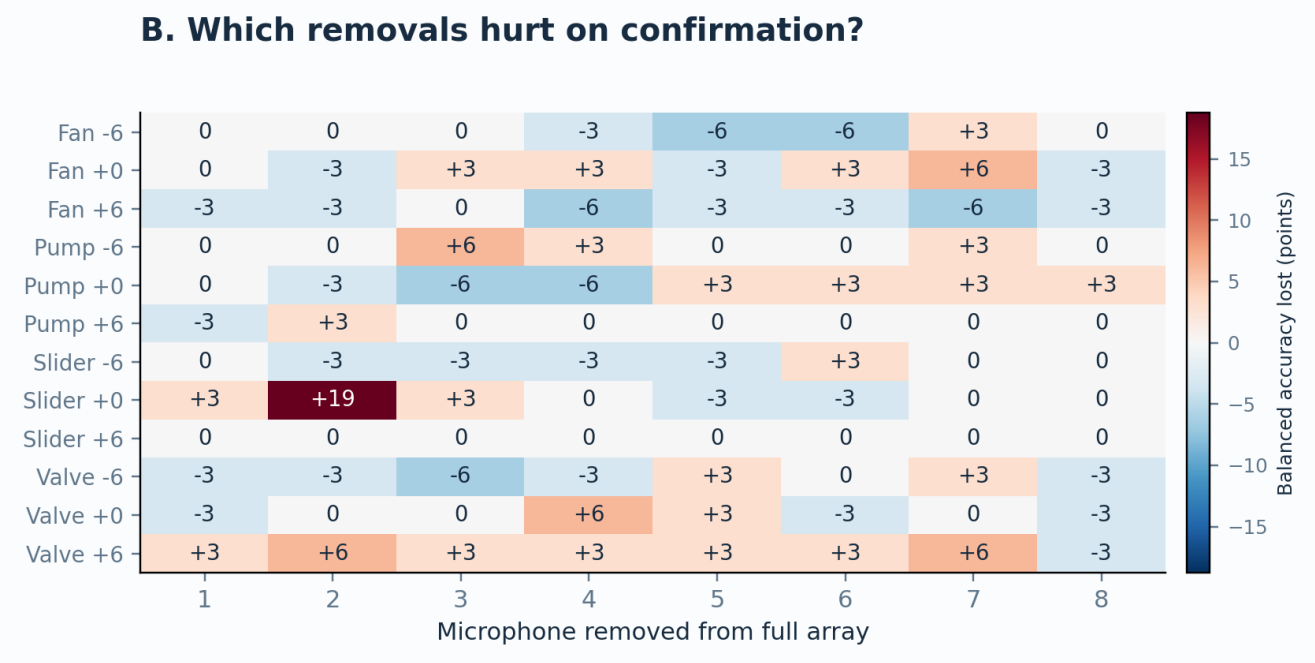


All singleton controls were declared in advance. Mic 1 is the representative baseline. Reporting all controls prevents a favorable reduced panel from hiding a better simple microphone.

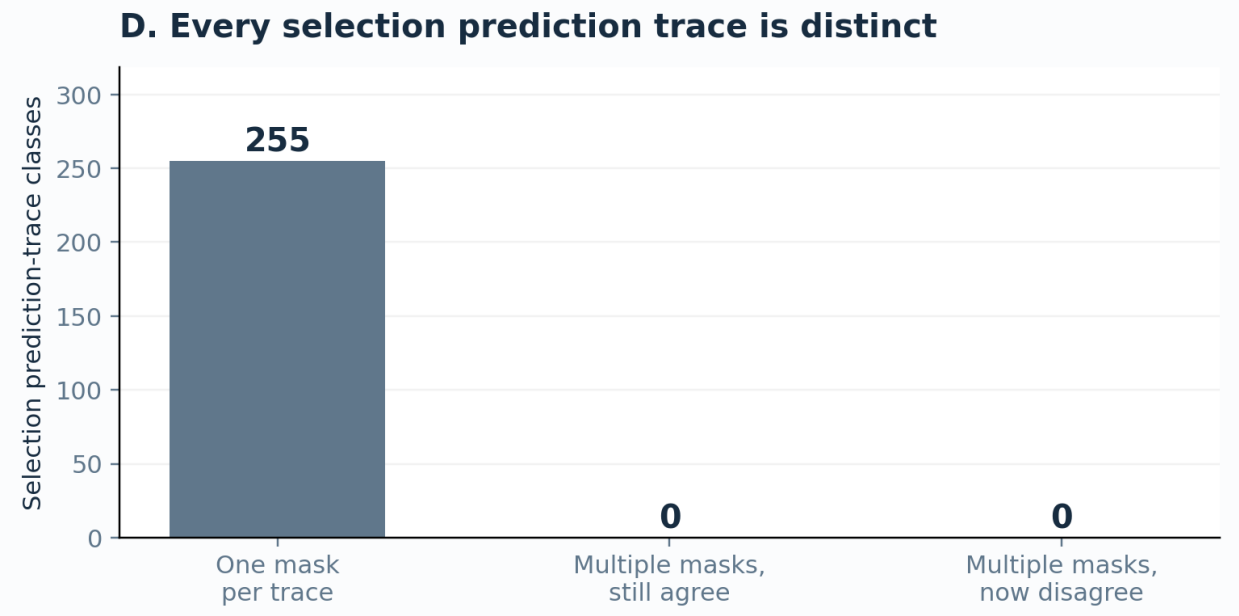
Protected type	Full array BA	Nominal BA	Normal recall	Abnormal recall
Fan	51.04%	52.08%	50.00%	54.17%
Pump	54.17%	54.17%	41.67%	66.67%
Slider	50.00%	44.79%	47.92%	41.67%
Valve	56.25%	50.00%	60.42%	39.58%

The nominal-panel class recalls remain visible even when an aggregate accuracy looks favorable. These numbers describe one protected machine per type, with all noise mixtures grouped by stem.

Channels contribute different information



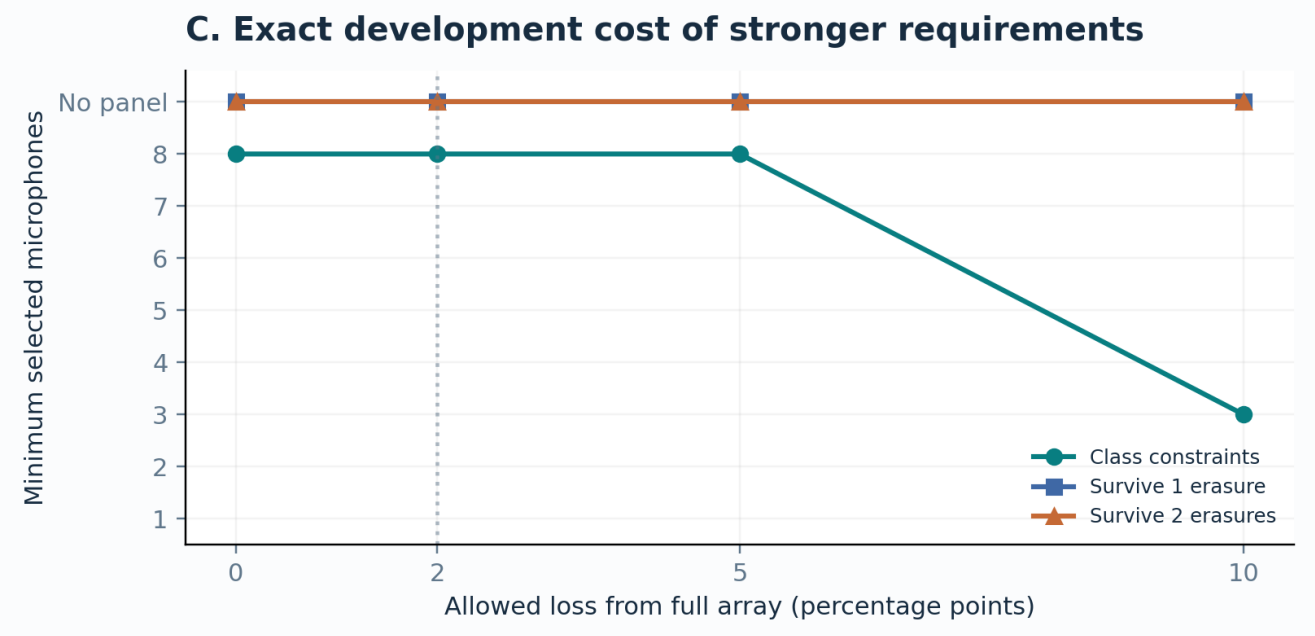
Positive values are balanced-accuracy points lost when removing a microphone from the full array; negative values mean removal improves this fixed classifier. SNR labels identify nominal archive mixture conditions, not measured per-recording ratios.



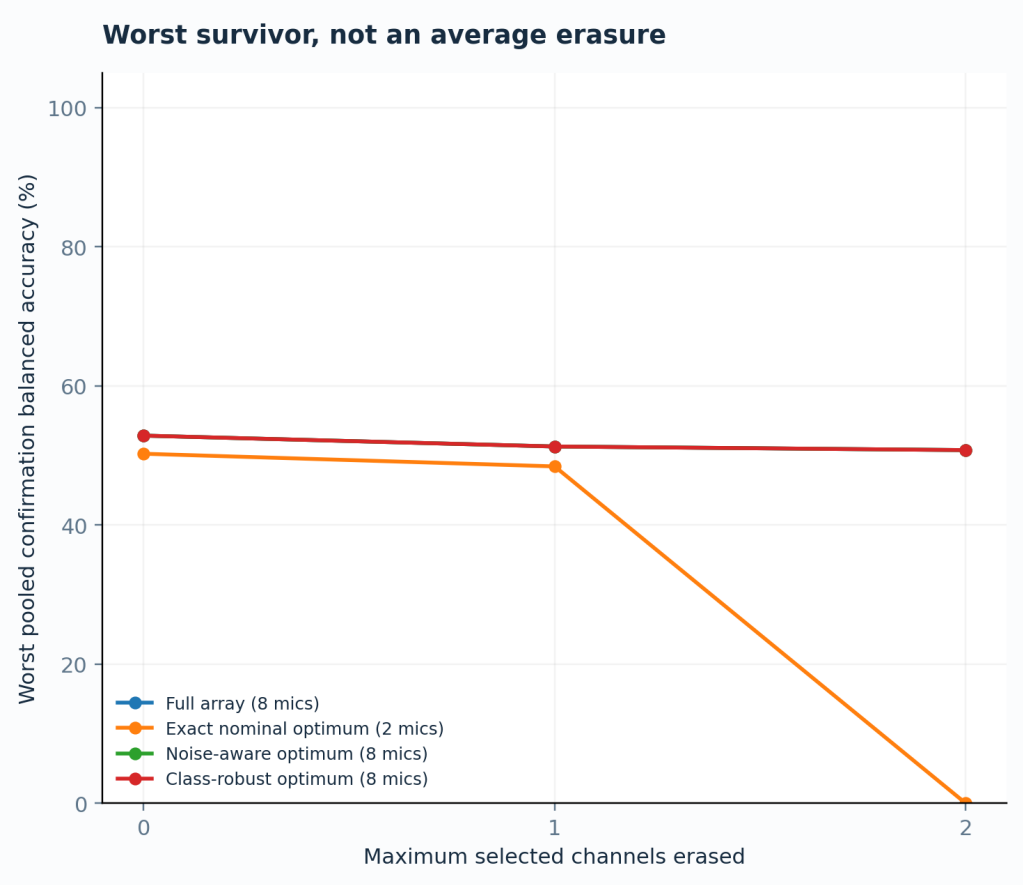
All 255 masks have different complete selection prediction traces. No empirical alias class of distinct panels occurs, so this study supplies no alias-splitting witness on confirmation.

Equal complete selection traces are empirical aliases. A split on confirmation does not establish acoustic equality or physical nonidentifiability.

The cost of surviving missing channels



The frontier is selected entirely from development data. No panel means all 255 nonempty possibilities fail the stated requirement. Costs cannot fall when an erasure requirement becomes stronger.



The confirmation curve uses the worst survivor after at most the specified number of lost channels. An empty survivor has no diagnostic input. This is a descriptive stress test, not a hardware failure-rate estimate.

What is certified - and what is measured

A complete finite optimization certificate

Every nonempty subset of eight microphones has one mask in 1 through 255. Given the saved numerical predictions, class recall and balanced accuracy are rational counts. Checking every mask and ordering feasible panels by channel count, then integer mask, proves the stated finite optimum or infeasibility. It does not prove that this empirical table represents every physical operating condition.

Erasures are a different question from multicover

The fused nearest-reference classifier can get worse after adding a channel. Its erasure certificate therefore checks every survivor explicitly. The existing Singularity robust-coverage engine is reused separately for per-channel separation of empirical normal/abnormal descriptor boxes. The multicover theorem applies to that fixed separation relation. Box overlap means its sufficient certificate fails; it is not proof that physical conditions are indistinguishable. The empirical development-box test finds 12 of 12 type-by-noise class distinctions without any individually separating microphone. Its complete separation requirement is infeasible at erasure budgets 0, 1, 2. This is a limit of the conservative empirical-box certificate, not proof of physical indistinguishability.

Physical provenance and the protected split

The official MIMII public 1.0 release supplies ten-second, eight-channel recordings at 16 kHz and sixteen bits. Its -6/0/+6 dB archive labels are nominal mixture conditions based on model-average signal power, not measured per-recording SNR guarantees. The deterministic subset has 512 recording stems and 1,536 noise mixtures across four types and four machine IDs. IDs 00/02 train, 04 selects and 06 confirms. Corresponding stems retain all three noise mixtures together. Hidden factory-noise source identities and temporal sessions are unavailable, so their independence is unverified.

The protocol was externally frozen before signal acquisition in repository commit 30132a5f7e332c832dc7ad471937a97b3483aa6f. Selective ZIP64 retrieval acquired 2,852,963,714 compressed audio bytes and 3,932,282,880 raw WAV bytes. The source manifest preserves original identities, archive offsets, lengths, CRC32 and compressed/raw SHA-256 hashes. Whole-archive MD5 values remain publisher metadata rather than verified checksums of a complete download. A second raw-feature route uses Python WAV decoding and an explicit NumPy FFT/Welch implementation. It verifies all selected members and reproduces all 159,744 descriptors to maximum absolute discrepancy 1.4433e-15, within 2e-11 tolerances.

The paired bootstrap resamples whole stems with every mixture together, stratified by type and class. Its intervals are conditional on four fixed protected machines. They do not estimate uncertainty over new machines. This balanced supervised identity-transfer stress test is distinct from the original normal-only per-ID anomaly benchmark.

Source attribution and reproducibility

Harsh Purohit and colleagues, Hitachi, Ltd. (2019). MIMII Dataset: Sound Dataset for Malfunctioning Industrial Machine Investigation and Inspection. DOI 10.5281/zenodo.3384388. CC BY-SA 4.0. Official record: https://zenodo.org/records/3384388. Primary paper: https://arxiv.org/html/1909.09347v1. The retained repository package includes protocol, source receipts, features, all-mask predictions, certificates, sensitivity analyses and an independent verifier. The earlier MAFAULDA studies remain unchanged.